Joshua Zhanson

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Employment

Microsoft *Mountain View, CA*

Applied Scientist 2 | Microsoft PowerPoint *October 2022 - Present*

- **Powerpoint Agent Mode**: Developed orchestrator prompt and tool prompts for agentic workflow. Created procedures to ensure quality of fast-changing prompt with many upstream and downstream stakeholders.
- **Visual creation in M365 Copilot Chat:** Led integration of visual creation capabilities into M365 Copilot Chat and Microsoft Visual Creator Copilot extension with **5K DAU**. Evaluated later model upgrade to GPT-4o, increasing image generation satisfaction by **26.4%** and overall chat satisfaction by **3.6%**; a new model architecture led to a **3.63%** increase in thumbs up rate, **13.62%** increase when image skill triggered, and **5.4%** increase in positive feedback. Adding style dimensions increased thumbs up rate by **3.78%** and achieved ChatGPT parity.
- **Artifact prompting architecture and automatic LLM prompt optimization**: Designed LLM prompting architecture and evaluation pipeline for diffusion-based artifact creation. Applied automatic prompt optimization tool to several Designer artifacts, facilitating prompt iteration and model upgrades, and achieving **76.47%** improvement over baseline for invitations artifact and reducing problematic generations by **47%**.
- **Template retrieval system**: Improved template retrieval search using a joint transformer + TF-IDF + multimodal / LLM architecture, increasing **precision by 93%** and **recall by 115%.** Evaluated using offline crowdlabeling and online A/B flighting, increasing design success rate by **9.13%**, seen rate by **3.78%**, and kept rate by **16.83%**.

Education

Carnegie Mellon University School of Computer Science *Pittsburgh, PA*

Master of Language Technologies *August 2022*

QPA: 3.80/4.00 Advisor: Yonatan Bisk Supported by **NSF Graduate Research Fellowship**

Carnegie Mellon University School of Computer Science *Pittsburgh, PA*

Bachelor of Science in Computer Science, Minor in Machine Learning *May 2020*

QPA: 3.95/4.00 Dean's List: Fall 2016 - Spring 2019 College & University Honors

Senior thesis: *Investigating and Robustifying Proximal Policy Optimization*

Advised by Emilio Parisotto, Adarsh Prasad, and Ruslan Salakhutdinov

Research Projects

Learning Visual Representations through Embodied Interaction Exploration *August 2020 - July 2022*

- Created Find One and Interaction Exploration environments in **Python** built on AI2THOR interactive embodied household robotics simulator to explore visual representation learning with embodiment
- Designed customizable ResNet visual encoders and decoders and LSTM policy model architectures in **Pytorch** for control with pixel inputs and outputs and designed a self-supervised policy and visual pretraining task
- Implemented custom variants of reinforcement learning algorithms Advantage Actor-Critic and Proximal Policy Optimization with hogwild asynchronous multiprocess training to allow running 8+ parallel environments
- Built multiprocess autoencoder baseline, supervised topline, and visual probe experiment pipeline to evaluate quality of learned representations on datasets with 2M+ images generated from different heuristic agent policies in AI2THOR simulator

On Proximal Policy Optimization's Heavy-tailed Gradients *August 2019 - May 2020*

- Accepted to **ICML 2021**

Proprioceptive Spatial Representations for Generalized Locomotion *June 2018 - July 2019*

- Accepted to **Workshop on Structure & Priors in Reinforcement Learning** at **ICLR 2019**

Skills

LLM frameworks: LLM prompt engineering ~ Agentic tools & orchestration ~ Evaluation & LLM-as-a-judge

Languages & technologies: Python ~ C/C++ ~ Pytorch ~ Numpy ~ Pandas ~ Git <3

Last updated February 28, 2026